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When Does a Physical Interaction Become a Valid Registered Measurement?

A VVV-QMRF Registration-Layer Framework with the K9_E Class C Testable Hypothesis and an Experimental Specification for Extended Wigner’s Friend

Working Paper v3.0 (*promoted from draft 2026-05-28 — all phases P1-P7 complete*) **Author:** Viet Nguyen Xuan (VietVunVut) **Affiliation:** Independent Researcher, Vietnam **Contact:** viet@vvvqmrf.com | <https://vvvqmrf.com> **Repository:** <https://github.com/AIhugART/VVV-QMRF-VietVunVut-Quantum-Measurement-Registration-Framework> **Date:** 2026-05-28 **Status:** Working paper. All formal claims are Class D (proposed) or Class C (conjecture/qualified) unless stated otherwise. Critique is explicitly invited. **Plan reference:** [papers/paper_003/VVV-QMRF_Working_Paper_v3.0_plan.md](#) **Base version:** v2.0 ([papers/Testable_Prediction_Section/.../VVV-QMRF_Working_Paper_v2.0.md](#))

DISCLAIMER: VVV-QMRF is independent personal research, not Standard Quantum Mechanics, not peer-reviewed, and not experimentally confirmed. K9_E (P9) is classified **Class C (qualified)**: structurally testable but empirically unconfirmed. Evidence is real but ambiguous; noise as an alternative explanation cannot be ruled out (v30 noise sensitivity analysis FAIL). Confirmation or rejection requires a purpose-designed K9-S12 photonic Extended Wigner’s Friend experiment. VVV-QMRF does not replace Standard QM, revise the Born rule, or invoke consciousness. Full boundary protocol: [DISCLAIMER.md](#). Formal definitions: [documents/research_documents/project_vvv_qmrf_class_c/06_references/VVV-QMRF_Definitions.m](#)

Abstract

Standard quantum mechanics specifies, via postulate P3, that a measurement of observable A on state $|\psi\rangle$ yields eigenvalue a_k with probability $|\langle a_k|\psi\rangle|^2$. What P3 does not specify is *when* a physical interaction constitutes a valid registered measurement event. This silence generates the von Neumann chain problem — every measuring apparatus becomes entangled with the system it measures, with no postulate indicating where this chain terminates — and leaves the Heisenberg cut formally undefined.

This paper proposes VVV-QMRF (VietVunVut Quantum Measurement Registration Framework): a registration-layer extension of quantum mechanics grounded in Buddhist Pramāṇa epistemology (Dignāga–Dharmakīrti tradition). The framework introduces a K-side registration space K, structurally separate from the physical Hilbert space H ($K \neq H$), and proposes a six-condition test for valid registered measurement. These six conditions formalize when a physical interaction crosses from the ρ -side into the K-side as a self-certified, validity-bearing registration event. Applied to Extended Wigner’s Friend (EWF) scenarios, this yields the K-side incommensurability conjecture: $K_F \perp K_W$ holds when a joint registration space K_{joint} is structurally required but cannot satisfy all six conditions simultaneously for both observers.

The K-side registration space is axiomatized via eight frozen registration-logic axioms (K1–K8,

Layer 1) and nine updatable bridge theorems (T1–T9, Layer 2). K1 defines the K-state carrier set (five-tuple: M, o, cert, t, V); K2 imposes a strict total temporal order; K3 formalizes self-certification; K4 assigns default validity; K5 governs invalidation and cross-registration contradiction; K6 defines cross-registration authority; K7 formalizes registration process closure (provisional $\rightarrow$ final validity); K8 guarantees validity preservation under cross-space embeddings. Layer 2 extends this structure with bridge theorems T1–T9, including K5_prospective (a conservative pre-instantiation evaluation mode required for probability assignment), K7_trace (a closure transition record), and D_enc (a transition-encoding semantic definition). T4-H — the N-observer colimit existence hypothesis — has been verified as a full theorem across all four steps (RCA 4.74/5, 2026-05-28), upgrading the three-observer prediction to Class C.

K9_E is the probability postulate (Postulate P9) motivated by the K-space architecture. It is not a theorem derivable from K1–K8 alone — the core axioms define structural properties (registration, validity, incommensurability) but do not uniquely determine a probability rule. K9_E fills this gap as P9: $P(o|K) = \text{Tr}(E_o \rho) \cdot (1 - \beta \cdot K_{\text{ctx}})$, where $\beta \in [0,1]$ is a single free parameter and K_{ctx} is the fraction of contextual observer registrations in prospective incommensurability with the current registration. At $\beta = 0$, K9_E reduces exactly to the Born rule. Six of the eight terms in K9_E are entirely new concepts not present in Standard QM. The proposed experimental specification (K9-S12 Modified Bong Protocol — single waveplate at angle $\alpha = 31^\circ$) predicts Genuine LF $1 = +0.0891$ (8.6σ), $\delta\langle A_1 B_2 \rangle = -0.0355$ (20.8σ), and figure of merit $\text{FOM} = 8.6$; this proposal was submitted to arXiv as a separate experimental child paper (paper_002, 2026-05-27).

Empirical status (Class C qualified): A genuine non-circular fit to raw Proietti et al. (2019) data yields $\beta = 0.598$, visibility $V = 0.939$, $\Delta\chi^2 = 5.35$ (2.31σ) in favor of K9_E over quantum mechanics with uniform visibility. However, a v30 noise sensitivity analysis returned FAIL: the noise threshold for evidence robustness is 0.10σ RMS (well below the 3.0σ threshold for PASS). Random noise at any magnitude produces $\Delta\chi^2 \geq 5.35$ in approximately 50% of realizations due to K9_E’s directional sensitivity and the small sample of four data points. The A0B0 setting alone drives 80% of the $\Delta\chi^2$ improvement. The 2.31σ signal reflects model flexibility, not confirmed physical suppression. Adversarial tests (4/4 PASS) and operationalizability gates (3/3 PASS) confirm K9_E’s structural robustness, but empirical confirmation remains open and requires the K9-S12 photonic EWF experiment with dedicated noise characterization.

This paper presents three logically independent projects under a one-way motivation chain. **Project A** (BE $\leftrightarrow$ QM Comparative Mapping) provides a 30-node, 39-edge comparative framework between Buddhist Pramāṇa epistemology and quantum measurement — an interpretive framework that motivated the registration-layer architecture but does not itself constitute evidence for that architecture. **Project B** (VVV-QMRF Conceptual Framework) develops the formal K-space axiomatization (K1–K8, Layer 1 frozen) and bridge theorem suite (T1–T9, Layer 2 updatable), including the conjectured structure-preserving map $\varphi: K \rightarrow B(H)$ with necessary conditions N_1 – N_T (Class D supporting conjecture). **Project C** (K9_E Testable Hypothesis) is the scientifically falsifiable claim: the K9_E probability postulate (P9) produces predictions structurally different from Standard QM ($\delta_S \neq 0$ when $\beta > 0$), avoids the Frauchiger–Renner paradox via K5 V_{prov} mechanism, and is falsifiable via the K9-S12 Modified Bong Protocol. $A \rightarrow B \rightarrow C$ is a one-way motivation chain, not a logical derivation. A null K9_E result falsifies Project C but does not invalidate Projects A or B.

VVV-QMRF does not replace Standard Quantum Mechanics. It does not revise postulates P1–P4 or the Born rule (K9_E recovers Born exactly at $\beta = 0$). It does not invoke consciousness. It does not claim Buddhist epistemology proves quantum mechanics. All formal claims are classified by evidence level; classification criteria are stated in the project’s formal definitions document.

Falsification requires a photonic EWF experiment with the K9-S12 single-waveplate protocol or equivalent. Critique is explicitly invited.

1. The Registration Layer Gap

1.1 What Quantum Mechanics Specifies

Standard quantum mechanics is built on four postulates:

- **P1 (State):** A physical system is represented by a density operator $\rho \in D(H)$.
- **P2 (Observables):** Physical quantities are represented by self-adjoint operators A on H .
- **P3 (Measurement):** Measurement of A on state $|\psi\rangle$ yields eigenvalue a_k with probability $|\langle a_k | \psi \rangle|^2$, after which the state updates to the corresponding eigenstate.
- **P4 (Dynamics):** Between measurements, the state evolves unitarily via the Schrödinger equation.

P3 specifies what outcome appears and with what probability. P3 does not specify when a physical interaction counts as a measurement, what structural conditions distinguish a measurement from a non-measurement interaction, or what stops the measurement chain. This is the registration layer gap.

1.2 The Von Neumann Chain Problem

Von Neumann (1932) observed that if apparatus $A1$ measures system S , $A1$ becomes entangled with S . If apparatus $A2$ then measures $A1$, $A2$ becomes entangled with $A1+S$. No postulate of standard QM specifies where this chain terminates. Decoherence (Zurek 2003) explains why superpositions become effectively classical at the macroscopic scale, but it does not explain when a physical interaction constitutes a registration event in the sense of a definite recorded outcome. The von Neumann chain problem is a registration-layer problem, not a decoherence problem.

1.3 The Heisenberg Cut

Copenhagen quantum mechanics (Bohr 1935; Heisenberg 1958) assigns measurement authority to “the classical apparatus” without formally defining what makes an apparatus classical, why it has registration authority, or what structural conditions produce a definite outcome. VVV-QMRF addresses both problems by introducing a registration layer K , separate from the physical layer ρ , and proposing formal conditions for valid registration events within K . The K1–K8 axioms (§4) formalize this registration layer structurally.

2. The K-side Registration Space

2.1 Layer Separation

VVV-QMRF introduces a strict separation between two layers:

- **Physical layer (ρ -side):** Physical states, observables, and dynamics as described by Standard QM. This layer is not modified.
- **Registration layer (K-side):** The space of registration events, recording conditions, and validity states. $K \neq H$. K-side structure does not alter ρ -side dynamics.

2.2 Minimal K-state

A registration event is represented by a minimal K-state tuple:

$$k = \langle M, o, \text{cert}, t, V \rangle$$

The K-side registration space K is the collection of such tuples produced by a registering system R over time. Section §4 formalizes this minimal K-state as the K1–K8 axiom set.

2.3 Source: Buddhist Pramāṇa Epistemology (Project A)

The registration-layer architecture is structurally derived from Buddhist Pramāṇa epistemology (Dignāga, 5th century; Dharmakīrti, 7th century). The structural extraction concerns: **Svasaṃvedana** (self-certifying cognition, source for K3), **Arthakriyā** (validity as functional success, source for K4/K7), and **Anumāna/Pratyakṣa** (inference and perception as distinct registration modes). This is Project A: a comparative interpretive framework. Project A motivates Project B via one-way structural inspiration; it does not constitute empirical evidence for the K-space architecture or K9_E.

3. The Valid Registered Measurement Test

3.1 The Six Conditions

Interaction X is a valid registered measurement for registering system R if and only if:

- | | | |
|----------------------------|---|--------------|
| Condition 1 (Physical): | X occurs at the physical ρ -side. | |
| Condition 2 (Admission): | X is admitted into K-side as M_X for system R . | |
| Condition 3 (Process): | $M_X \in R$ ordered by registration time. | [K2, E6] |
| Condition 4 (Self-cert): | $\sigma_R(M_X) = 1$ intrinsically within K_R . | [K3, E1] |
| Condition 5 (Validity): | $V(M_X) = 1$ by default upon admission. | [K4, E7] |
| Condition 6 (Non-invalid): | No later M' revises $V(M_X)$ to 0. | [K5, K7, E7] |

The cert and V fields are formalized as K3 (self-certification) and K4/K5/K7 (default validity and invalidation) in §4.

3.2 The K-side Stopping Proposition

Proposition (Class D proposed): If Conditions 1–6 hold for M_X , no further K-side registration act is required to certify M_X . The von Neumann registration regress terminates at M_X on the K-side without modifying ρ -side dynamics. Condition 4 (self-certification) provides the formal property that Copenhagen assigns to “classical apparatus” without definition.

3.3 Observer-Indexed Self-Certification

For multi-observer scenarios: $\sigma_R(M) = 1$ iff M occurred as a K-side registration event of R , determined intrinsically within K_R , not by any $M' \neq M$ and not by any $R' \neq R$. The independence $\sigma_F(M_F) \perp \sigma_W(M_W)$ is formalized in K3 (§4.1).

4. K-Space Architecture: Axioms K1–K8 and Layer 2

This section presents the formal axiomatization of the K-side registration space. The axioms and bridge theorems are the structural backbone of VVV-QMRF and the foundation on which K9_E (§5) and the EWF incommensurability result (§6) rest.

Scope boundary (v2.4 §0.6 audit): The K-space axiomatization in this section is **purely structural**. It contains zero probability equations, zero numerical values, and zero experimental data. K9_E (§5) is a separate probability postulate (P9) that is **not derivable from K1–K8 alone** — the axioms define structural properties (registration, validity, incommensurability) but do not uniquely determine a probability rule. Data fitting (§7) is a separate empirical analysis. This three-layer separation is architectural: K1–K8 (structure) $\rightarrow$ K9_E (postulate) $\rightarrow$ data fitting (empirical).

The authoritative peer-synced source is `documents/research_documents/meta_architecture/K_Space_Axiomatization` (canonical) and `documents/research_documents/project_vvv_qmrf_class_c/01_axiomatization/K_Space_Axiomatization` (working copy), both v2.4.

4.1 Layer 1 (Frozen): Axioms K1–K8

Layer 1 contains eight registration-logic axioms whose text is frozen unconditionally. K1–K4 and K8 carry no Level 4 semantic dependencies. K5, K6, and K7 have conditional semantic dependencies on Level 4 scope (when they fire / when closure occurs) but their text is frozen regardless of Level 4 changes.

Axiom	Name	Core function	BE lineage	Class
K1	Carrier Set	K_R = set of 5-tuples $k = \langle M, o, \text{cert}, t, V \rangle$; $\text{cert} = 1$ for all $k \in K_R$ (admission invariant); t-injectivity: distinct events have distinct timestamps	Pramāṇa — cognition as structured event: act, object, self-awareness, result, validity	C
K2	Temporal Order	(K_R, <_R) is strict total order (chain): $k_1 <_R k_2$ iff $t(k_1) < t(k_2)$; discrete: no K-side registration identity between consecutive events	Kṣaṇabhaṅgavāda — momentariness; registration time is discrete within K_R	D

Axiom	Name	Core function	BE lineage	Class
K3	Self-Certification	cert(k) = $\sigma_R(M) \in \{0,1\}$ determined intrinsically within K_R; independent of any $R' \neq R$; observer-indexed independence	Svasaṃvedana — self-certifying cognition: a cognition certifies its own occurrence without a second-order cognition	D
K4	Default Validity	$V(k) = 1$ upon instantiation for $\neg \text{isNull}(k)$; $V(k)$ $= 0$ for null events ($\text{isNull}: o$ $= \emptyset \wedge \Delta I = 0$); V is provisional until K7 closure	Svataḥ prāmāṇya — — intrinsic validity by default; arthakriyā (causal efficacy)	D
K5	Invalidation	$V(k_1) \rightarrow 0$ iff $\exists k_2$ later in K2 order, $k_2 \perp k_1$ in shared C_K , k_2 has $\text{Auth}(K6)$; fires only when requires_K_joint $= 1$; pre-closure reversible, post-closure absolute	Parataḥ prāmāṇya — invalidity detected extrinsically; bādhaka pramāṇa	D
K6	Cross-Reg. Authority	$\text{Auth}(k_2 \rightarrow k_1,$ $C_K) = 1$ iff: both in same C_K ; $V(k_2) = 1$; $k_1 \in$ $\text{scope}(D_joint)$; non-transitive across distinct C_K	Bādhaka pramāṇa — a contradicting cognition must itself be a valid pramāṇa	D

Axiom	Name	Core function	BE lineage	Class
K7	Closure	R closes at t_close when no pending requires_K_joint demands remain; V_prov $\rightarrow$ V_final; post-closure: no new k, K5 irreversibility absolute, no new D_joint	Niścaya (ascertainment) — a cognition becomes determinate at cognitive closure	D
K8	Embedding Preservation	For embedding i: K_R $\rightarrow$ K_X: V_X(i(k)) = V_R(k) at t_embed; all five fields preserved; post-embedding V evolves under K_X's K4–K7	Anugama — continuity: validity accompanies a cognition into a broader context	D

Note on K1 claim class: K1 is classified C because the K-state tuple definition is shared with the K9_E postulate (which is Class C); K1 provides K9_E's domain. K2–K8 remain Class D as structural axioms without empirical content.

4.2 K5_prospective: Conservative Pre-Instantiation Evaluation Extension

K9_E requires assessing incommensurability *before* an outcome is registered — to compute the suppression factor $f_{\text{perp}}(o) = P(\text{K5 fires prospectively on hypothetical outcome } o)$. K5's standard post-hoc mode (modifying V of an actual tuple) is not designed for this. K5_prospective is a **conservative extension** that adds a new evaluation target while preserving K5's identical conditions (i)–(iii):

K5 (post-hoc): $V(k_1) \rightarrow 0$ iff $\exists k_2: k_1 <_R k_2 \wedge k_2 \perp k_1 \wedge \text{Auth}(k_2 \rightarrow k_1)$
Target: actual $k_1 \in K_R$. Effect: V_prov modified.

K5_prospective: K5 fires on k_{o*} iff $\exists k_{\text{prev}}: k_{\text{prev}} <_{\text{joint}} k_{o*}$
 $\wedge k_{o*} \perp k_{\text{prev}} \wedge \text{Auth}(k_{o*} \rightarrow k_{\text{prev}})$
Target: hypothetical $k_{o*} = \langle M^*, o, \text{cert}=1, t^*, V=1 \rangle$.
Effect: contributes to $f_{\text{perp}}(o)$ in K9_E only.

Same conditions (i)–(iii). Same structural logic. Different target and effect.
K5_prospective does NOT modify V of any actual tuple in K_R. It has no operational role outside K9_E. This conservative extension was upgraded from “semantic extension” to an explicit axiom-level clause in v29 (2026-05-23, 3-Round RCA Round 2 score 4.90/5), resolving assumption [A-E1] through the T9 construction theorem.

Claim class: C (conservative extension of K5; same conditions; new evaluation mode only).

4.3 Layer 2 (Updatable): Bridge Theorems T1–T9

Layer 2 bridge theorems connect K1–K8 to the framework’s operational definitions (Level 4 predicates: D_joint, requires_K_joint, Bridge_EWF, ODC_K). Layer 2 is updatable independently of K1–K8.

Theorem	Name	Core function	Class	Status
T1	K_joint Construction	K_joint exists as categorical colimit of embedding diagram when requires_K_joint = 1; order = transitive closure of embedded orders + cross-temporal relations	D	Pending L4 freeze
T2	$\perp_K$ Derivation	K_A $\perp_K$ K_B iff requires_K_joint = 1 $\wedge$ no admissible K_joint; traced to K5 conflict (V_prov forced to 0 in candidate K_joint) or K7 lock	D	Pending L4 freeze
T3	Bridge_EWF	Bridge_EWF = 1 derivable from K5 in EWF: M_F registers definite o_F; M_W registers same lab as superposition; no reinterpretation preserves both without changing validity claim; conditional on AJVS	D/C	Pending L4 freeze

Theorem	Name	Core function	Class	Status
T4	N-Observer Colimit	$K_joint(R_1, \dots, R_N)$ exists as colimit for $N \geq 2$ when pairwise $AdmJoint +$ global commutativity hold; $\perp_K$ non-transitive in general		T4-H THEOREM 4/4 (2026-05-28)
T5	K_joint Associativity	$K_joint(K_joint(A, \mathbb{B}), C)$ $\cong$ $K_joint(A, B, C)$ as K1–K8- preserving isomorphism; conditional on T4-H		T4-H resolved; L4 freeze pending
T6	Decoherence Update	Decoherence- induced ρ -side change produces K-side registration-state update when Conditions 1–6 hold	D	Updatable
T7	IRB Scope Propagation	Intrinsic Relational Binding (entanglement) between R_i and R_j induces $requires_K_joint$ $= 1$ under specific conditions	D	Updatable
T8	$K5_prospective$ Frequency Bridge	$f_perp =$ $E[I(K5_prospective$ $fires)] =$ fraction of K_ctx with $prospective \perp$; derives $K9_E$ suppression term structurally; eliminates assumption [A-E2]	C	Structural derivation from $K5p$

Theorem	Name	Core function	Class	Status
T9	K_ctx Construction	K_ctx = { $\varphi_{ij}(k_j) \in$ K_joint : $k_j \in$ K_{R_j}, requires_K_joint = 1, temporally compatible}; φ_{ij} = canonical K8-constrained T1 embedding; eliminates [A-E1] via 5 lemmas (L1-L5)	C	L1–L5 proven

T4-H THEOREM (4/4, 2026-05-28, RCA 4.74/5): All four steps verified: - Step 1: $C_{\{K\text{-space}\}}$ satisfies category axioms (identity, composition, associativity). - Step 2: $K_{\text{colim}} = (\coprod_i K_i)/\sim$ constructed; all five fields well-defined via lexicographic t-assignment and embedding-time V snapshot; 5/5 verification gates PASS. - Step 3: K1–K8 preserved through quotient construction via T-PRES Lemma + T-REP Corollary (K2 acyclicity resolved) + V monotone dynamics. - Step 4: Universal property verified: $u([k,i]) := f_i(k)$ is well-defined (diagram compatibility + T-REP), K8-preserving, and unique.

T4-H is no longer a hypothesis. T4 conclusions hold for all $N \geq 2$. The three-observer prediction (§10) is upgraded from Class C-conditional to Class C. T1 ($N=2$ constructive) remains independently valid without invoking the colimit universal property.

4.4 K7_trace and D_enc: Canonical Layer 2 (v2.4, 2026-05-27)

K7_trace — Closure Transition Record: A conservative extension of K7. Defines $\Delta_{\text{closure}}(k) := V_{\text{prov}}(k) - V_{\text{final}}(k)$ at t_{close} . Records the transition magnitude (0 or 1) without modifying V or creating new tuples. Claim class: C-canonical. Promoted from BB-VVV local origin to canonical Layer 2 by RCA 4.77/5.

D_enc — Transition-Encoding Registration Act: A semantic definition (parent: K7_trace). $\text{Enc}(M_{\text{aware}}, k_F) = 1$ iff $o(M_{\text{aware}})$ depends counterfactually on $\Delta_{\text{closure}}(k_F)$. Pattern: same as K5_prospective (binary classification of hypothetical act). Claim class: C-canonical.

Four canonical consumers of K7_trace (cross-validation of architectural status):

Consumer	Usage	Status
T_BB V3 Step 1 (BB-VVV)	Δ_{closure} as V_{prov} substitute post-closure (Baumann-Brukner 2024)	§9.1 — T_BB Class C conditional
D_enc	Parent for Enc predicate	Canonical Layer 2
3-OBS hierarchical registration transition	T4-H THEOREM unlocks Class C upgrade for 3-observer prediction	§10 — Class C (2026-05-28)

Consumer	Usage	Status
FR-VVV avoidance chain Step 2	Frauchiger-Renner K5→K6→V=0 path; K7_trace scenario-agnostic across B&B and FR	§9.2 — V_FR2 PASS (2026-05-28)

The four-consumer canonical status confirms K7_trace is an architectural element, not a single-use construct.

4.5 K-Space and Buddhist Pramāṇa: One-Way Motivation, Not Empirical Evidence

The BE lineage entries in §4.1 record the structural inspiration for each axiom. This is Project A’s role in VVV-QMRF: a one-way motivation from Pramāṇa theory’s analysis of valid cognition to K-space’s analysis of valid registration.

What this does NOT mean: - Buddhist epistemology does not prove K1–K8. - K1–K8 are not derived from Buddhist texts; they are derived from the registration-layer problem posed by Standard QM. - Project A’s 30-node, 39-edge BE↔QM comparative mapping is a separate independent project (see §12.3 for the A→B→C independence statement). - A result that falsifies K9_E does not falsify Buddhist epistemology, the BE↔QM mapping, or the K-space architecture.

5. K9_E: Probability Postulate (P9)

ERRATUM note (from Phase 8 source document, 2026-05-23): This section was formerly titled “K9_E Formal Derivation.” K9_E is **NOT derived from K1–K8**. It is a **POSTULATE** — a probability assignment rule motivated by K-space structure ($\perp_K$, K_ctx) but not uniquely determined by K1–K8 axioms. This distinction is load-bearing: the entire §5 must be read as proposing P9, not proving it.

5.1 Why K1–K8 Alone Do Not Determine a Probability Rule

The eight axioms of Layer 1 define the structural properties of the K-side registration space: what tuples are admitted (K1), how they are ordered (K2), how self-certification works (K3), how default validity is assigned (K4), how cross-registration contradiction fires (K5), what counts as authority for invalidation (K6), when the registration process closes (K7), and how validity is preserved under embedding (K8).

Nowhere in K1–K8 is there a formula that maps K-state tuples to real-valued probabilities. The axioms determine:

- *Which* registration events are valid (cert = 1, V = 1)
- *When* one event invalidates another (K5 conditions)
- *How* events embed across K-spaces (K8)

They do not determine *how much probability* to assign to each outcome. Multiple distinct probability functions are consistent with the K1–K8 structure. In particular, the Born rule $P(o) = \text{Tr}(E_o \rho)$ is consistent with K1–K8 (with $\beta = 0$, the K-space constraints are satisfied trivially). But so is

any function that respects the V-gate ($V = 0 \rightarrow$ no probability), the isNull-gate ($\text{isNull}(k) \rightarrow$ no probability), and normalizes over the outcome space.

K9_E fills this gap by adding a specific probability assignment rule as **Postulate P9**, motivated by the K-space incommensurability structure (K5_prospective, T8, T9) but not derivable from it. This is analogous to how the Born rule P3 in Standard QM is an irreducible postulate of the theory — it is not derivable from P1 (state), P2 (observables), or P4 (dynamics).

5.2 The K9_E Equation

Postulate P9 (K9_E Probability Postulate):

$$P(o \mid k_i, \text{Exp}) = \frac{\text{Tr}(E_o \rho_i) \cdot [1 - \beta \cdot f_{\text{perp}}(o, k_i, K_{\text{ctx}})]}{Z_E(k_i)}$$

where:

$$f_{\text{perp}}(o, k_i, K_{\text{ctx}}) = \frac{|\{k_j \in K_{\text{ctx}} : \text{K5_prospective fires on } k_o^* \text{ vs } k_j\}|}{|K_{\text{ctx}}|}$$

$$K_{\text{ctx}}(k_i, \text{Exp}) = \{\varphi_{ij}(k_j) \in K_{\text{joint}} : k_j \in K_{\{R_j\}}, \text{requires_K_joint}(R_i, R_j) = 1, \text{temporally compatible with } k_i\} \quad [\text{T9}]$$

$$Z_E(k_i) = \sum_{\{o'\}} \text{Tr}(E_{\{o'\}} \rho_i) \cdot [1 - \beta \cdot f_{\text{perp}}(o', k_i, K_{\text{ctx}})] \quad [\text{normalization}]$$

$$\beta \in [0, 1) \quad \text{-- single free parameter (suppression strength)}$$

Born rule limit (exact, not approximate):

K9_E reduces to the Born rule exactly under any of the following equivalent conditions:

$P(o|k) = \text{Tr}(E_o \rho)$ iff any of:

- (i) $\beta = 0$ [suppression OFF -- K-space has no effect on P]
- (ii) $K_{\text{ctx}} = \emptyset$ [no contextual observers -- f_{perp} undefined $\rightarrow 0$ by convention]
- (iii) $f_{\text{perp}}(o) = 0 \forall o$ [all outcomes compatible -- no prospective K5 firing]
- (iv) Single observer $N=1$ [implies $K_{\text{ctx}} = \emptyset \rightarrow$ (ii)]

For condition (i): $\beta = 0 \rightarrow [1 - 0 \cdot f_{\text{perp}}] = 1$ for all $o \rightarrow Z_E = \sum \text{Tr}(E_o \rho) = 1 \rightarrow P = \text{Tr}(E_o \rho)$. Standard single-observer laboratory experiments have $N = 1$ (condition iv) and therefore K9_E is observationally equivalent to Standard QM in all non-EWF contexts.

5.3 Eight-Term Provenance: What Is New in K9_E

K9_E assembles eight structural components. Six of these eight terms are entirely new concepts not present in Standard QM. This table records the provenance of each term — it is a traceability record, not a derivation proof.

#	Term	Definition	Source	In Standard QM?
T1	$\text{Tr}(E_o \rho_i)$	Born rule probability (POVM formulation)	Standard QM	[YES]
T2	β	Suppression strength, $\beta \in [0,1)$	Free parameter — the single adjustable parameter of K9_E; analogous to coupling constants (α , G, g) in physics	[NO] NEW
T3	$f_{\text{perp}}(o, k_i, K_{\text{ctx}})$	Fraction of contextual registrations prospectively incommensurable with outcome o	K5_prospective + T8 — structural derivation from binary K5/K6 primitives (T8-H1: fraction form is uniquely forced)	[NO] NEW
T4	$C(o_i, o_j)$	Compatibility map: 1 if outcomes are QM-orthogonal (incompatible), 0 otherwise	K-space construction from ρ_{joint} at setup, stored as K-side lookup	[NO] NEW
T5	$K_{\text{ctx}}(k_i, \text{Exp})$	Set of contextual K-states from other observers via T9 morphism + temporal compatibility	T9 + K2 — K_{ctx} is a theorem, not an assumption (L1–L5 in T9)	[NO] NEW
T6	$Z_E(k_i)$	Normalization: $\Sigma_{\{o'\}} \text{Tr}(E_{\{o'\}} \rho_i) \cdot [1 - \beta \cdot f_{\text{perp}}(o_i, k_i, K_{\text{ctx}})]$	Modified from QM (Standard QM) auto-normalizes; K9_E suppression breaks auto-normalization, requiring explicit Z)	[WARN] MODIFIED

#	Term	Definition	Source	In Standard QM?
T7	$V(k) = 0 \rightarrow \text{no } P$	Bhrānti gate: invalid registrations receive no probability	K4 + K5 $\rightarrow$ PP-1 v2 — grounded in BE concept bhrānti (erroneous cognition)	[NO] NEW
T8	$\text{isNull}(k) \rightarrow \text{no } P$	Anupalabdhi gate: null events ($o = \emptyset, \Delta I = 0$) receive no probability	K4 isNull guard — grounded in BE concept anupalabdhi (non- apprehension)	[NO] NEW

Summary: Of eight terms in K9_E: 1 from Standard QM (T1: Born rule), 1 modified from QM (T6: normalization), **6 entirely new** (T2–T5, T7, T8). The four original K9_E construction assumptions ([A-E1]–[A-E4]) are fully resolved: [A-E1] ELIMINATED (T9, 5 lemmas), [A-E2] ELIMINATED (T8/T8-H1), [A-E3] RECLASSIFIED as free parameter β , [A-E4] BE-anchored. **Net: 0 orphaned assumptions, 1 free parameter (β).**

5.4 K9_E Is a Postulate, Not a Theorem

This is the most important boundary statement in this section.

K9_E is **Postulate P9** — a proposed probability assignment rule added to the VVV-QMRF framework alongside K1–K8. It is not derivable from K1–K8 alone. The axioms define structural properties (registration, validity, incommensurability) but do not uniquely determine a probability rule.

The relationship between K1–K8 and K9_E is the same as the relationship between P1, P2, P4 (state, observables, dynamics) and P3 (Born rule) in Standard QM: the structural postulates define the mathematical space, but the probability rule is an additional postulate with its own empirical motivation.

What K1–K8 do: Define the K-space structural constraints (carrier, order, certification, validity, invalidation, authority, closure, embedding). Supply the structural vocabulary used in K9_E (K5_prospective, T8, T9). Constrain the form of a probability rule (V-gate via T7, isNull-gate via T8), but do not determine β or the specific suppression formula.

What K9_E adds: A specific probability assignment rule $P(o|K)$ with a single free parameter β . The suppression term f_{perp} is structurally motivated by K5_prospective incommensurability and formally derived (via T8, T9) to have the fraction form — but the *presence* of the suppression term (i.e., that $\beta \neq 0$ in nature) is a postulate, not a derivation.

Consequence: K9_E is falsifiable ($\beta = 0$ recovers Born exactly; any measured $\beta \neq 0$ confirms K9_E; K9-S12 protocol provides the definitive test). K1–K8 are not individually falsifiable — they are structural definitions of the registration layer.

5.5 Distinguishability from Standard QM

When $\beta > 0$, $K9_ctx \neq \emptyset$, and f_perp is outcome-dependent, $K9_E$ produces predictions structurally different from Standard QM.

Distinguishability condition: $\delta P(o) = P_{\{K9E\}}(o) - \text{Tr}(E_o \rho) \neq 0$ iff ALL of:

- (I) $\beta > 0$
- (II) $K_ctx \neq \emptyset$ (requires $K_joint = 1$ for at least one observer pair)
- (III) $\exists o, o' : f_perp(o) \neq f_perp(o')$

Condition (III) is critical: if f_perp is constant across outcomes, the suppression factor $[1 - \beta \cdot f_perp]$ is a uniform multiplier that cancels in the normalized ratio $P/Z_E \rightarrow \text{Tr}(E_o \rho)$. This is the PP-2 v2 cancellation insight: $K9_E$ must suppress *differentially* across outcomes to produce an observable signal.

In EWF scenarios with Bell-State Measurement (BSM), the compatibility map $C(o_F, o_W)$ is outcome-dependent (some (o_F, o_W) pairs are compatible, others are incommensurable), producing outcome-dependent f_perp and therefore a genuine $\delta P \neq 0$.

Theoretical magnitude (CHSH, $\beta = 0.5$): $\delta_S = -0.055$. This is a *theoretical* value — the predicted deviation in the CHSH parameter for Proietti-type 2-observer EWF. At $\beta = 0.9$, $\delta_S = -0.179$ ($\approx 2.4\sigma$ detection horizon). No empirical detection of δ_S has been achieved to date. See §7 for empirical status.

3-observer amplification (Class C, T4-H THEOREM): The 3-observer K_ctx is larger, producing an amplification factor of approximately $11\times$ relative to 2-observer. Predicted $\delta_M3 = -0.223$ at $\beta = 0.3$ (illustrative, conditional only on $K9_E$ postulate P9 — T4-H is now THEOREM, §10).

5.6 Projects B and C Are Independent Claims

The φ -map conjecture (§11, Project B) and $K9_E$ (Project C) are logically independent:

Project B (φ -map): Conjectures the existence of a structure-preserving map $\varphi: K \rightarrow B(H)$. Derives necessary conditions N_1 – N_T . Provides φ -conditional scope boundaries for existing QM interpretations. This conjecture is Class D and concerns the *formal relationship* between K-space and the algebra of bounded operators. It does not make testable predictions about measurement statistics.

Project C ($K9_E$): Proposes a probability postulate P9 with a single free parameter β . Makes testable predictions (K9-S12). Can be confirmed or falsified by experiment. $K9_E$'s empirical status is entirely independent of whether φ exists.

Independence in both directions: - A proof that $\varphi: K \rightarrow B(H)$ exists (or does not exist) would not confirm or falsify $K9_E$. - A K9-S12 experimental confirmation of $K9_E$ ($\beta \neq 0$ measured) would not prove φ exists. - A K9-S12 null result ($\beta = 0$ measured) would falsify $K9_E$ but would not falsify the φ -map conjecture or the K-space architecture.

This independence is architectural: Project B concerns the structural embedding of K into H ; Project C concerns the probability assignment within K . They share the K-space vocabulary (K1–K8, T1–T9) but their claims are logically disjoint.

6. Extended Wigner’s Friend and K-side Incommensurability

6.1 The Extended Wigner’s Friend Setup

In the Extended Wigner’s Friend scenario (Frauchiger and Renner 2018; Brukner 2018), two observer pairs interact with a shared quantum system:

- **Friend F** measures quantum system S inside a sealed laboratory. From F’s perspective, a definite outcome o_F is recorded: $\sigma_F(M_F) = 1$.
- **Wigner W** models the entire laboratory (F + S) as a unitarily evolving quantum state, then performs an interference measurement M_W on F’s laboratory.

Standard QM describes both perspectives as valid without providing a formal account of their structural incompatibility. This is where the registration layer offers new structure.

6.2 Applying the Six Conditions to Each Observer

Friend F:

$X_F \rightarrow M_F \in R_F$ [Condition 3, K2]
 $\sigma_F(M_F) = 1$ [Condition 4, K3]
 $V_F(M_F) = 1$ [Condition 5, K4]
No M' contradicts M_F [Condition 6, K5/K7]
 $\rightarrow M_F$ is a valid registered measurement within K_F .

Wigner W:

$X_W \rightarrow M_W \in R_W$ [Condition 3, K2]
 $\sigma_W(M_W) = 1$ [Condition 4, K3]
 $V_W(M_W) = 1$ [Condition 5, K4]
No M' contradicts M_W [Condition 6, K5/K7]
 $\rightarrow M_W$ is a valid registered measurement within K_W .

Both observers satisfy the six conditions independently within their own K-sides. The question is whether a joint registration space K_{joint} can contain both K_F and K_W as jointly valid entries.

6.3 The requires_ K_{joint} Predicate

Definition (admissible joint K-side registration space, Class D proposed):

$\text{AdmJoint}(K_{\text{joint}}; A, B) = 1$
iff there exist embeddings $i_A: A \rightarrow K_{\text{joint}}$ and $i_B: B \rightarrow K_{\text{joint}}$ such that:
(i) embeddings preserve act, outcome, cert, registration time/order, validity;
(ii) self-certification remains intrinsic to each embedded act;
(iii) Conditions 1-6 remain satisfied for each embedded structure;
(iv) no required registration-state update in K_{joint} invalidates either embedded structure while both are still claimed as jointly valid;
(v) K_{joint} does not import an external certifier as source of self-certification.

Definition (requires_ K_{joint} predicate, Class D proposed):

$\text{requires}_{K_{\text{joint}}}(A, B) = 1$
iff A and B are each valid or provisionally valid within their own K-side
AND A and B are brought under a shared validity demand D_{joint}

AND D_joint requires both to be assessed as parts of the same registration target,
 history, counterfactual claim, or validity claim
 AND truth of D_joint cannot be evaluated while leaving A and B in fully independent K-sides
 AND preserving D_joint requires a K_joint in which A and B are jointly valid.

requires_K_joint(A, B) = 0

iff no shared D_joint is imposed, or D_joint can be evaluated without embedding A and B
 into one candidate K_joint.

Operational sufficient conditions for requires_K_joint = 1:

- **Condition A (Wigner interference):** W performs an interference measurement on F's lab; M_W registers a superposition description while M_F registers a definite outcome for the same lab. $\rightarrow$ requires_K_joint = 1.
- **Condition B (Direct comparison):** F and W directly compare records and a logical contradiction is detectable. $\rightarrow$ requires_K_joint = 1.
- **Condition B2 (LF constraint):** An LF inequality requires F-side and W-side claims to have simultaneous cross-observer validity. $\rightarrow$ requires_K_joint = 1.

Operational sufficient conditions for requires_K_joint = 0:

- **Condition C (No interference, no comparison):** W does not perform interference on F's lab. $\rightarrow$ requires_K_joint = 0.
- **Condition D (Separable state):** Shared state is separable; M_F and M_W act on non-overlapping subsystems. $\rightarrow$ requires_K_joint = 0.
- **Condition E (Independent bookkeeping):** Records stored together but no joint validity demand imposed. $\rightarrow$ requires_K_joint = 0.

6.4 Formal Definition of K-side Incommensurability

****Definition (K-side incommensurability relation $\perp_K$, Class D proposed):****

A $\perp_K$ B

iff requires_K_joint(A, B) = 1

AND there exists no admissible K_joint such that:

- (i) A and B both embed into K_joint,
- (ii) their respective self-certifications are preserved,
- (iii) their respective validity conditions remain satisfied, and
- (iv) no required registration-state update in K_joint invalidates either A or B while both are still claimed as jointly valid.

Definition (K-side comparison context C_K): A minimal shared frame in which two registration acts are evaluated for compatibility. C_K requires: (a) both acts admitted into the same comparison domain; (b) both indexed to the same registration target; (c) comparison does not presuppose joint validity. C_K is strictly weaker than AdmJoint.

Definition (cross-registration authority): In C_K , M_later has valid cross-registration authority w.r.t. M_earlier iff: (a) both belong to same C_K ; (b) $V(M_later) = 1$; (c) M_later's content directly concerns the same registration target as M_earlier; (d) the comparison architecture does not arbitrarily privilege one observer.

Boundary clauses: - $\perp_K$ does not assert that either physical event fails to occur on the ρ -side. -

$\perp_K$ does not mean either observer's outcome is false within its own K-side. - $\perp_K$ is not equivalent to a registration-null event $\text{Null}_K(e)$. - $\perp_K$ applies only when both sides are valid/provisionally valid within their own K-sides.

6.5 K_{joint} Failure and K-side Incommensurability

Bridge lemma (Class D proposed; application status: Class C):

$\text{Bridge_EWF}(D_{\text{joint}}; M_F, M_W) = 1$

iff D_{joint} requires F-side and W-side registrations to be evaluated as jointly valid parts of one laboratory registration history
AND M_F registers a definite friend-side outcome o_F
AND M_W registers the same laboratory as coherent superposition (no definite o_F as W-side)
AND LF/no-go comparison requires both claims to support one cross-observer validity constraint
AND no reinterpretation inside K_{joint} preserves both without changing the validity claim of at least one side.

$\text{Bridge_EWF}(D_{\text{joint}}; M_F, M_W) = 1 \rightarrow M_W \perp M_F$.

Relativization defense (rejected): A relativized K_{joint} hosting only meta-descriptions (“within K_F, M_F registered $|h\rangle$ ”) does not satisfy D_{joint} . D_{joint} requires joint validity of the *original* registration claims, not meta-descriptions of them. Relativizing the contents abandons D_{joint} rather than satisfying it.

Conditional lemma (K_{joint} failure, Class D proposed): Under $\text{requires_}K_{\text{joint}} = 1$, $\text{Bridge_EWF} = 1$, and valid cross-registration authority, no admissible K_{joint} exists such that $\sigma_F(M_F) = 1$, $\sigma_W(M_W) = 1$, $V(M_F) = 1$, and $V(M_W) = 1$ simultaneously hold. $K_F \perp_K K_W$ follows. **Claim class of Step 4:** C/D boundary — conditional on full Bridge_EWF semantic proof.

6.6 The Falsifiable Prediction and ODC_K

VVV-QMRF conjectures that configurations satisfying $\text{requires_}K_{\text{joint}} = 1$ are the natural candidates for LF-level violation when the quantum state is sufficiently entangled.

Operational data criterion (ODC_K , Class C proposed):

$\text{ODC}_K(\text{Data}, \text{Cfg}) = K_{\text{joint_fails}}$

iff Cfg satisfies $\text{requires_}K_{\text{joint}} = 1$ via D_{joint} and Bridge_EWF ,
AND no joint registration model J_K satisfying:
(i) jointly valid F-side and W-side registrations,
(ii) Conditions 1-3, 5, 6 preserved (Condition 4 by construction),
(iii) AdmJoint condition (iv): no required invalidation while both are jointly claimed,
(iv) observed probability distribution reproduced within predeclared tolerance τ ,
(v) not reclassifying a Bridge_EWF config as mere independent bookkeeping.

τ must be fixed before data collection; ODC_K is sensitive to the choice of τ .

Compatibility checks with existing data:

Proietti et al. (2019): The three correlators with $\text{requires_}K_{\text{joint}} = 1$ ($\langle A_1 B_1 \rangle$, $\langle A_1 B_0 \rangle$, $\langle A_0 B_1 \rangle$) contribute positively to the violated expression $S = 2.407$; the one term with $\text{requires_}K_{\text{joint}} = 0$

($\langle A_0 B_0 \rangle$) is subtracted. This allocation is structurally compatible. The VVV-QMRF reading is a term-role interpretation of an aggregate inequality, not a per-configuration experimental confirmation.

Bong et al. (2020): Two-regime structure:

μ regime	Reported result	VVV-QMRF reading
$\mu = 0.80-0.81$	Bell non-LF violated; LF inequalities NOT violated	Regime 1: requires_K_joint = 1 active, entanglement insufficient for LF failure
$\mu \approx 0.87$	First LF violation (Semi-Brukner)	Transition Regime 1 $\rightarrow$ 2
High μ	All categories violated including Genuine LF	Regime 2: K_joint failure exposed

Per-facet μ -threshold ODC_K prediction:

Facet class	$\mu=0.80-0.81$	High- μ
Bell non-LF	K_joint_fails (violated)	K_joint_fails (violated)
Brukner	K_joint_exists (not violated)	K_joint_fails (violated)
Semi-Brukner	K_joint_exists (not violated)	K_joint_fails (violated)
Genuine LF	K_joint_exists (not violated)	K_joint_fails (violated)

The prediction hierarchy (Bell non-LF first, then Semi-Brukner, then Genuine LF) matches the observed μ -threshold ordering. **Not yet a facet-level confirmation** — full ODC_K stage 3 model-fit test has not been performed. These are compatibility checks.

6.7 Connection to K1–K8 and K5_prospective

The Level 4 formalism of §6.1–6.6 can now be traced directly to Layer 1 axioms via the K-space architecture (§4):

requires_K_joint = K5 precondition:

```
requires_K_joint(A, B) = 1
   $\iff$  K5 firing precondition met: comparison context C_K exists
   $\iff$  D_joint imposes a shared validity demand on K_F and K_W
      (K5 formal block: "fires only when requires_K_joint = 1")
```

K5 does not fire in isolation — it fires only when there is a joint validity demand. The requires_K_joint = 0/1 distinction in §6 is precisely the K5 precondition distinction in Layer 1. Every operational sufficient condition A–E for requires_K_joint maps to whether K5's precondition is met.

Bridge_EWF = K5_prospective evaluation:

```
Bridge_EWF(D_joint; M_F, M_W) = 1
   $\iff$  K5_prospective fires on hypothetical k_o* vs k_prev = k_F:
    -- k_F <_joint k_o*           [K2 temporal order in K_joint from T1]
    -- k_o*  $\perp$  k_F within C_K    [K5 registered contradiction, minimal  $\perp$  definition]
    -- Auth(k_o*  $\rightarrow$  k_F, C_K) = 1 [K6 authority]
```

Bridge_EWF = 1 means exactly that K5_prospective would fire if W’s hypothetical registration k_o^* were realized — the structural bridge between Level 4 formalism and Layer 1 axioms. This is the connection exploited by T8 to derive f_{perp} (§5.3).

K7_trace and K_joint failure: When $K_F \perp_K K_W$ is established, K7_trace records $\Delta_{\text{closure}}(k_F) := V_{\text{prov}}(k_F) - V_{\text{final}}(k_F)$. The four canonical consumers of K7_trace (§4.4, §9.3) all derive from this same closure event in structurally distinct EWF scenarios.

7. Empirical Status (Class C Qualified)

Classification: K9_E = **Class C (qualified)** — structurally testable, empirically **UNCONFIRMED**. Evidence is real (a genuine non-circular fit shows 2.31σ improvement over QM-uniform-visibility) but ambiguous (noise at any magnitude produces an equivalent improvement in ~50% of realizations). Confirmation or rejection requires the K9-S12 purpose-designed photonic EWF experiment.

7.1 D1 Proietti: Genuine Non-Circular Fit

Version history (critical): The Phase 10 data fitting document (2026-05-23) identified a *circular fit* error in the earlier Phase 10 analysis: the “data” used were reconstructed as $E_{\text{exp}} = V_{\text{exp}} \cdot E_{\text{QM}}$, mathematically guaranteeing $\beta = 0$ as the best-fit result (a tautology, not an empirical finding). The v29 analysis (2026-05-23) replaced this with a **genuine non-circular fit** using raw correlator values extracted directly from Proietti et al. (2019) Figure 3 (Wigner_figure_3.md SOT document).

Genuine fit results (raw Proietti Figure 3 data):

Quantity	Value	Meaning
β (best-fit)	0.598	K9_E suppression parameter at genuine fit
V (visibility, fitted)	0.939	Higher than circular-fit $V = 0.854$; non-uniform visibility detected
χ^2/DOF	0.670 (DOF = 2)	Good fit quality, $p = 0.51$
$\Delta\chi^2$ (K9_E vs QM-uniform-V)	5.35	K9_E improves fit over QM with uniform visibility
σ -significance	2.31σ	Marginal detection level

Why the raw data differs from reconstructed data: The raw Proietti Figure 3 correlator values (e.g., $A0B0 = -0.678$) differ substantially from reconstructed values (-0.604 in the circular-fit analysis). The non-uniform pattern across the four correlators is what drives the genuine-fit improvement — when K9_E’s directional suppression is applied, it preferentially reduces some correlators (those with high $\text{requires_K_joint} = 1$ contribution) over others ($\text{requires_K_joint} = 0$), producing a better fit than a uniform-visibility model.

Two K9_E model variants co-exist with different calibration paths: - **Additive model** (utils/k9e_predictor.py): $E = E_{\text{QM}} \cdot [1 - \beta \cdot n_{\text{BSM}} \cdot g_{\text{ctx}}]$, $g_{\text{ctx}} \approx 0.039$ (from theoretical $\delta_S = -0.055$ at $\beta = 0.5$). - **Multiplicative model** (proietti_raw_fit.py): $E = E_{\text{QM}} \cdot$

$[1 - \beta \cdot g_{\text{eff}}]^{\wedge n_{\text{BSM}}}$, $g_{\text{eff}} = 0.146$ (from PP-4 sanity check). Produces larger suppression; used for genuine fit.

The two models agree at first order in $\beta \cdot g$ but diverge for $\beta > 0.3$.

7.2 v30 Noise Sensitivity Analysis: FAIL

The v30 noise sensitivity analysis (P10-NOISE, 2026-05-24) asked: *How large must non-uniform noise be to produce $\Delta\chi^2 \geq 5.35$ by chance?*

Result: FAIL. The noise threshold is 0.10σ RMS — far below the 3.0σ threshold required for PASS.

Metric	Value	Interpretation
Noise threshold (2σ , B4 Monte Carlo)	0.10σ RMS	Noise at ANY magnitude produces $\Delta\chi^2 \geq 5.35$ in $\sim 50\%$ of realizations
PASS threshold	3.0σ RMS	Threshold for “noise ruled out”
Verdict	FAIL ($0.10 \ll 3.0$)	Non-uniform noise cannot be ruled out as explanation
Single-setting fragility (B2)	1.85σ at A0B0	Only a 1.85σ shift at A0B0 eliminates K9_E advantage
A0B0 contribution to $\Delta\chi^2$	80%	Nearly the entire K9_E “signal” is driven by one data point

What this means: The 2.31σ signal reflects **model flexibility** (K9_E has one more degree of freedom than QM-uniform-visibility), not confirmed physical K9_E suppression. With only 4 correlator values and K9_E’s directional sensitivity, random noise at any magnitude produces a $\Delta\chi^2 \geq 5.35$ improvement approximately half the time. The genuine fit improvement is real but not distinguishable from a noise artifact with current data.

Why this downgraded K9_E from “genuine” to “qualified”: The v29 upgrade to Class C (genuine) was based on three conditions: (1) genuine non-circular fit, (2) K5_prospective axiom upgrade, (3) T4-H Step 1. Conditions (2) and (3) remain valid. Condition (1) still holds as a genuine fit, but the noise analysis reveals that the fit evidence is insufficient to rule out noise. The v30 P10-NOISE FAIL downgraded the empirical sub-condition from “real evidence” to “real but ambiguous evidence.” Class C structural qualification is unchanged; the (genuine) $\rightarrow$ (qualified) downgrade concerns the empirical confidence level only.

Methodology rationale (conservative): The noise analysis was designed to ERR ON THE SIDE OF CAUTION — a Type I error (falsely claiming noise is ruled out) is fatal to project credibility; a Type II error (correctly noting that noise cannot be ruled out) is recoverable via the K9-S12 experiment.

7.3 K9E-PAT: CLOSED as UNRESOLVABLE

Open item K9E-PAT asked whether the K9_E multiplicative suppression pattern (predicted 2BSM/1BSM ratio ≈ 2) is confirmed by Proietti data.

Result: CLOSED as UNRESOLVABLE (v31 RCA 4.92/5).

The empirical 2BSM/1BSM ratio of -0.78 is not a meaningful test of the K9_E pattern:

1. The ratio -0.78 is computed from $A0B1 + A1B0$ avg = -0.0235 divided by $A1B1 = +0.0179$ — both are sub- σ residuals (consistent with zero).
2. The ratio of two near-zero numbers carries no physical information about suppression structure.
3. Both K9_E model variants predict suppression ratio ≈ 2 : additive (2.000 exactly), multiplicative (1.913). The predicted value ≈ 2 is not discriminating for current data quality.
4. P10-NOISE already established that 4 data points are insufficient to test any pattern — A0B0 alone drives 80% of $\Delta\chi^2$.

The path to pattern confirmation is K9-S12: an alpha-sweep protocol (QWP angle 0° – 90°) with $N = 91,000$ events and dedicated noise characterization, providing direct 2BSM/1BSM comparison with sufficient statistics.

7.4 Adversarial Tests (4/4 PASS) and Operationalizability Gates (3/3 PASS)

Despite the empirically qualified status, K9_E passes all structural validity tests:

Adversarial tests (4/4 PASS — Phase 9): 1. Born limit recovery: $\beta = 0 \rightarrow P = \text{Tr}(E_o \rho)$ exactly (verified analytically and numerically). 2. Single-observer reduction: $N = 1 \rightarrow K_ctx = \emptyset \rightarrow K9_E = \text{Born}$ (verified). 3. V-gate (Bhrānti): $V = 0$ events receive no probability (verified). 4. isNull-gate (Anupalabdhi): null events ($o = \emptyset$) receive no probability (verified).

Operationalizability gates (3/3 PASS — G1/G2/G3, all 5.0/5): - G1: K9_E is operationally distinguishable from Standard QM ($\delta_S \neq 0$ when $\beta > 0$ and $K_ctx \neq \emptyset$): PASS. - G2: K9_E has a predeclared falsification condition (K9-S12 protocol, §8): PASS. - G3: K9_E's free parameter β is measurable in principle (Proietti-type EWF experiment): PASS.

Important boundary: Adversarial tests and operationalizability gates confirm that K9_E is *structurally consistent and testable*. They do not confirm it is empirically correct. The 4/4 PASS result establishes that K9_E is a well-formed scientific hypothesis; the P10-NOISE FAIL establishes that current empirical evidence is insufficient to confirm it.

8. K9-S12 Modified Bong Protocol

This section summarizes the experimental specification for the first dedicated test of the K9_E probability postulate. The full protocol is presented in the experimental child paper (paper_002, arXiv submitted 2026-05-27). Section §8 provides the VVV-QMRF-level framing; the photonic implementation details are in paper_002.

8.1 The Equatorial Gap: Why No Prior Test Exists

Proposition 1 (Equatorial Cancellation Theorem, paper_002 §3): Every published optical EWF experiment — Proietti et al. (2019) and Bong et al. (2020) — used Superobserver measurement settings at the Bloch sphere equatorial plane ($\theta = \pi/2$). At $\theta = \pi/2$, the squared basis overlaps $|\langle b|d \rangle|^2 = 1/2$ for all outcome pairs (b, d). This makes the overlap-dependent suppression factor f_perp constant across all outcomes — the suppression cancels in the normalization Z_E and

P_K9E reduces exactly to the Born rule. This is not a special property of K9_E; it holds for **any** overlap-only deformation of the form $P' = P_{\text{QM}} \cdot g(|\langle b|d \rangle|^2) / Z$.

Consequence: Both published optical EWF experiments are geometrically blind to the entire overlap-only class of deformations, including K9_E. No published experiment has varied the polar angle θ . The K9-S12 protocol is the first proposal to break this fixed point.

Signal structure: The K9_E signal $\delta\langle AB \rangle(\theta)$ vanishes identically at $\theta = \pi/2$ and is generically non-zero otherwise:

$$f_{\text{perp}}(+1, H) - f_{\text{perp}}(-1, H) = -\cos \theta \quad [\text{Eq. 11, paper_002}]$$

$$\begin{aligned} \text{At } \theta = \pi/2: \quad \cos(\pi/2) = 0 &\rightarrow f_{\text{perp}} \text{ constant} \rightarrow \delta\langle AB \rangle = 0 & [\text{equatorial null}] \\ \text{At } \theta = 31^\circ: \quad \cos(31^\circ) \approx 0.857 &\rightarrow f_{\text{perp}} \text{ outcome-dependent} \rightarrow & [\text{non-zero signal}] \\ &\delta\langle AB \rangle \neq 0 \text{ for } \beta > 0 \end{aligned}$$

This vanishing-at-equator structure is a **genuine observable** (Lemma 1, paper_002 §3.2): it cannot be eliminated by any unitary relabeling of the Superobserver’s measurement basis.

8.2 Single-QWP Design and Predictions

Experimental modification: A single quarter-wave plate (QWP) is re-inserted into the Bong et al. (2020) apparatus, tilting the Superobserver measurement from $\theta = \pi/2$ to $\theta = 31^\circ$. No other hardware change is required. $N = 91,000$ coincidences per setting (same as Bong 2020).

Why $\theta = 31^\circ$ (near-optimal): Grid search over $\theta \in [0^\circ, 90^\circ]$ shows $\theta = 35^\circ$ yields FOM = 8.8 vs. 8.6 at $\theta = 31^\circ$. The broad plateau means the exact optimum is non-critical: $\theta = 31^\circ$ is near-optimal, not uniquely optimal (paper_002 v94 update). The key criterion is that $\theta \neq \pi/2$ — any angle away from the equator probes the signal.

Quantitative predictions at $\theta = 31^\circ$ (exact numerical computation):

Observable	Prediction	Significance	Role
Gen LF 1 (Genuine LF Facet 1)	+0.0891	8.6σ	Confirms LF violation is preserved at $\theta = 31^\circ$
$\delta\langle A_1 B_2 \rangle = \langle A_1 B_2 \rangle_{\{\theta=31^\circ\}} - \langle A_1 B_2 \rangle_{\{\theta=\pi/2\}}$	−0.0355	20.8σ	Core K9_E signal — non-zero iff K9_E with $\beta > 0$
Minimum detectable β at 5 σ	0.07	—	Search sensitivity
Figure of Merit (FOM)	8.6	—	Combined LF + K9_E signal

Scope boundary (paper_002 §2.3): These predictions assume the overlap-only class (Level 0). The experiment tests whether nature exhibits overlap-dependent deformation; it does not test the full K9_E framework. Broader deformation classes (Level 1: density-matrix-dependent; Level 2: multi-partite; Level 3: non-geometric) lie outside Proposition 1’s scope and require independent experimental designs.

Loophole status: Under fair-sampling ($\eta \approx 0.87$), K9-S12 is a **loophole-open screening test**. A positive result ($\delta\langle A_1 B_2 \rangle \neq 0$ at $\theta = 31^\circ$) requires independent verification including θ -sweeps (paper_002 §8.2). A null $\delta\langle AB \rangle$ across a full θ -sweep would falsify the overlap-only class.

Falsification conditions: K9_E (overlap-only class) is falsified if $\delta\langle AB \rangle = 0$ at $\theta = 31^\circ$ within predeclared statistical tolerance, confirmed by θ -sweep.

8.3 Why IBM Quantum Was Rejected

The IBM Quantum approach to K9_E testing was evaluated and rejected (v31 RCA 4.92/5) on the basis of a **double category error**:

Error 1 (registration structure): K9_E requires a K-space registration structure — tuples $k = \langle M, o, \text{cert}, t, V \rangle$ with K3 self-certification ($\text{cert} = 1$ intrinsic), K4 default validity ($V = 1$ upon admission), K5 invalidation ($V \rightarrow 0$ via cross-registration contradiction), and K7 closure ($V_{\text{prov}} \rightarrow V_{\text{final}}$). Gate-model QPUs implement unitary circuits with binary qubit states. There is no concept of a K-side registration tuple, K_R carrier set, or validity field $V \in \{0,1\}$ on a QPU. K-space does not exist as a computational object on IBM Quantum hardware.

Error 2 (EWF scenario): K9_E specifically requires an Extended Wigner’s Friend configuration — a Friend observer registering a definite outcome within a sealed laboratory, and a Superobserver performing an interference measurement on the entire Friend+system composite. Gate-model circuits simulate quantum evolution but do not implement the structural architecture of multiple registering systems with K_joint construction and cross-registration authority. The “Friend” and “Superobserver” roles in K9_E are K-space structural roles, not qubit register labels.

Conclusion: K9_E’s testable predictions are predictions about photonic EWF statistics. Only an optical EWF apparatus implementing the K9-S12 protocol (or equivalent) constitutes a valid test platform.

8.4 Relationship to paper_002

paper_002 (“Have Optical Wigner’s Friend Experiments Been Blind to a Geometric Degree of Freedom?”) was submitted to arXiv 2026-05-27 as an independent experimental child paper. The VVV-QMRF framing in §8.1–8.3 provides the K-space motivation for the experimental protocol; the photonic implementation details, statistical analysis, and Supplemental materials are in paper_002.

paper_002 is self-contained as a physics paper: it presents the Equatorial Cancellation Theorem (Proposition 1) and single-waveplate protocol without requiring the VVV-QMRF K-space framework. The K9_E postulate (§5) provides one theoretical motivation for the overlap-dependent deformation parameter β ; paper_002 frames β as a phenomenological search parameter analogous to Standard Model Extension (SME) coefficients.

9. Connections to Baumann-Brukner and Frauchiger-Renner

This section documents how two specific papers in the EWF literature connect to VVV-QMRF’s K-space structure. These are **Layer 4 (Class D) structural compatibility assessments** — they show that VVV-QMRF’s machinery can be applied to the scenarios in these papers, not that the papers empirically confirm K9_E.

9.1 Baumann & Brukner (2024): T_BB Class C Conditional

Paper reference: Baumann, V. and Brukner, Č. (2024). “Wigner’s friend as a rational agent.” [arXiv reference; fit plan source: 09_Fitting_Baumann_Brukner/BB_VVV_fit_plan.md v1.4]

Scenario: A Friend F measures a quantum system inside a sealed lab; a Wigner W performs either a coherent (interference) measurement or a projective (read-out) measurement on the lab. B&B show that a rational agent (W) cannot consistently assign memory to F after a coherent measurement — the “no-awareness” result.

VVV-QMRF compatibility analysis (BB_VVV_compatibility_section.md v2.1):

Verification	Status	Finding
V1: K5 $\leftrightarrow$ B&B $q_{00} < 0$ condition	[WARN] PARTIAL (F4 triggered)	R_BB (near-readout, $x \approx 0$) $\neq$ R_K5 (interference, $x \approx \pi/4$). B&B’s no-valid-joint-model fires near readout; K5 fires at interference. Structurally different failure modes — not mathematically equivalent. F4 finding: at $x = \pi/4$ (maximum interference), $q_{00} = 0.5 > 0$ always, while requires_K_joint = 1 and K5 fires.
V2: K7 closure $\leftrightarrow$ B&B memory change Δp	[YES] PASS	K7 closure magnitude $\Delta p = 1 - 2\alpha^2 \cdot \sin^2(2x) / 2$ matches B&B memory change with identical functional form. Verified at 5 test points including readout ($x \approx 0$), maximal interference ($x = \pi/4$), and transition region.
T_BB (Option A)	[YES] Class C (conditional)	No-awareness result derived via K5 $\rightarrow$ K6 $\rightarrow$ V=0 chain using K7_trace + D_enc. G1 CLOSED by K7_trace §18 + D_enc §19. Computational verification PASS (bb_vvv_t_bb_verification.py).
T_BB’ (Option C)	[YES] CLOSED (superseded)	V1 F4 finding falsifies T_BB’ Step 1 premise (R_BB = R_K5 was required). T_BB (Option A) computationally verified; T_BB’ superseded.

P2-C ($\pi/8$ first-principles): The compatibility check at angle $x = \pi/8$ yields $\Delta p = |1 - 2\alpha^2| / 4$ exactly — a first-principles derivation from K7 closure (no free parameters). This is **exact** for all α^2

$\neq 0.5$ (symmetric state degeneracy at $\alpha^2 = 0.5$ confirmed as expected: $\Delta p = 0$).

T_BB derivation chain:

K5_prospective: W's coherent measurement $\rightarrow$ requires_K_joint(K_F, K_W) = 1
K7_closure: K_F closes at t_close(K_F)
K7_trace: $\Delta_{\text{closure}}(k_F) := V_{\text{prov}}(k_F) - V_{\text{final}}(k_F) \in \{0,1\}$
D_enc: Enc(M_aware, k_F) = 1 iff o(M_aware) counterfactually depends on Δ_{closure}
K5_fires: requires_K_joint(M_aware, M_W) = 1 $\rightarrow$ V(M_aware) $\rightarrow$ 0
Result: Friend F cannot retain awareness (V=0) $\rightarrow$ T_BB no-awareness result

Key boundary: V1's F4 finding does not invalidate T_BB. T_BB uses K7_trace + D_enc (Layer 2), not the R_BB = R_K5 equivalence. What F4 establishes is that B&B's no-valid-joint-model condition and VVV-QMRF's K5 incommensurability capture **structurally different aspects** of the Wigner's Friend problem — they are compatible frameworks analyzing different failure modes.

9.2 Frauchiger & Renner (2018): FR Paradox Avoided via K5 V_prov

Paper reference: Frauchiger, D. and Renner, R. (2018). “Quantum theory cannot consistently describe the use of itself.” *Nature Communications* 9, 3711.

Scenario: Four agents (F1, F2, W1, W2) in nested EWF setups. FR show that under three assumptions (Q: universality, C: consistency, S: single-outcome), agents reach mutually contradictory predictions. The contradiction requires that $V(k_{F1}) = 1$ AND $V(k_{W1}) = 1$ simultaneously — both the Friend's registration and the Wigner's registration are valid at the same time under the same cross-observer validity demand.

VVV-QMRF FR avoidance chain:

FR contradiction requires: $V(k_{F1}) = 1$ AND $V(k_{W1}) = 1$ simultaneously
 $\downarrow$
K5 fires at t_2 : $K_{F1} \perp_K K_{W1}$ within comparison context C_K
(W1's coherent measurement $\equiv$ requires_K_joint = 1)
 $\downarrow$
K6 Authority: Auth($K_{W1} \rightarrow K_{F1}$) = 1
 $\rightarrow V(k_{F1}) \rightarrow 0$ (K5 post-closure irreversible)
 $\downarrow$
Premise fails: $V(k_{F1}) = 1$ AND $V(k_{W1}) = 1$ NEVER simultaneously satisfied
 $\downarrow$
FR contradiction: AVOIDED -- K5 gates the joint validity condition at source

VVV-QMRF recast of FR assumption (S): The FR argument requires F1's outcome and W1's outcome to belong to the same validity domain (joint cross-observer fact). $K5 \perp_K$ prevents this: they are structurally incommensurable registrations. VVV-QMRF does not *reject* (S); it *scopes* (S) to within a single K-context. The FR contradiction never assembles within VVV-QMRF's structural machinery because the joint validity demand is blocked by K5 before it can be evaluated.

V_FR2 verification (script-verified, 2026-05-28): K7_trace records $\Delta_{\text{closure}}(k_{F1}) := V_{\text{prov}}(k_{F1}) - V_{\text{final}}(k_{F1})$ at W1's coherent measurement. This is **structurally identical** to T_BB Step 2 in B&B — the same K7_trace definition applies unchanged to a different scenario with different agents and a different contradiction mechanism. fr_vvv_k7trace_consumer_verification.py v1.0: OVERALL PASS.

T_FR status (Class D, blocked by G_FR2): A full T_FR (No-Joint-Validity Bridge Theorem) for the 4-agent scenario (F1, F2, W1, W2) requires the N=4 K_joint construction, which depends on T4-H for N=4. T4-H is THEOREM for N=2 (T1 constructive) and general N (abstract colimit proof, Steps 1-4). The N=4 concrete instantiation has not been explicitly verified. A simplified 2-agent version (F1+W1) is feasible and structurally established by V_FR2.

9.3 K7_trace + D_enc: Canonical Layer 2 with Four Consumers

The V_FR2 result confirms K7_trace as a **scenario-agnostic** Layer 2 primitive — it applies with identical definition across structurally distinct scenarios:

Consumer	Scenario	Role	Status
T_BB (BB-VVV)	Baumann-Brukner (2024): F + W angle sweep	V_prov substitute after K7 closure for T_BB no-awareness derivation	Class C conditional
D_enc	All scenarios with K7 closure	Parent: Enc(M_aware, k_F) = 1 iff counterfactual dependence on Δ_{closure}	Class C-canonical
3-OBS	Three-observer hierarchical EWF	Δ_{closure} propagation through $F1 \rightarrow F2 \rightarrow W$ hierarchy (§10)	Class C (T4-H verified)
FR (FR-VVV)	Frauchiger-Renner (2018): 4-agent nested EWF	V_FR2: $\Delta_{\text{closure}}(k_{F1})$ after W1's coherent measurement — blocks FR joint validity	Class D (T_FR blocked by G_FR2)

Architectural significance: K7_trace being consumed by four structurally distinct analyses (B&B angle-sweep, D_enc semantic, 3-OBS hierarchical, FR nested) confirms it is a legitimate Layer 2 primitive — not a single-use BB-VVV construct. This cross-validation was the primary criterion for canonical promotion (RCA 4.77/5, 2026-05-27).

10. T4-H THEOREM and 3-Observer Prediction

10.1 T4-H: From Hypothesis to Theorem (4/4 Steps, 2026-05-28)

T4-H is the N-observer Colimit Existence Theorem for the category $C_{\{K\text{-space}\}}$. It states: *for any finite diagram D of K-spaces with K1–K8-preserving morphisms, the colimit K_{colim} exists and satisfies K1–K8.*

T4-H was originally framed as a hypothesis (“T4-H”) because its proof required four sequential steps. As of 2026-05-28 all four are verified (RCA 4.74/5):

Step	Claim	Key technique	Status
Step 1	$C_{\{K\text{-space}\}}$ is a valid category	Identity morphisms + composition + associativity via K8-preserving maps	VERIFIED
Step 2	$K_colim = (\coprod_i K_i)/\sim$ is constructible	Lexicographic t-assignment (SP1), V-snapshot at embedding (SP2), transitive closure for $<_colim$ (SP3); 5/5 gates PASS	VERIFIED (RCA 4.73/5)
Step 3	K_colim satisfies K1–K8	T-PRES Lemma (all morphisms preserve t exactly, resolving K2 acyclicity); K3–K8 verified via quotient structure	VERIFIED (RCA 4.74/5, 2026-05-28)
Step 4	K_colim satisfies universal property	$u([k,i]) := f_i(k)$ well-defined (T-REP corollary), K8-preserving, unique	VERIFIED (2026-05-28)

T-PRES Lemma (key to Step 3): Any morphism $f: K_i \rightarrow K_j$ in $C_{\{K\text{-space}\}}$ satisfies $t_j(f(k)) = t_i(k)$ for all $k \in K_i$. This is because K8-preservation requires all five tuple fields ($M, o, cert, t, V$) to be preserved exactly. Consequence: all representatives of any equivalence class $[k, i]$ share the same t value (T-REP Corollary), ensuring K2’s acyclicity condition holds in K_colim without cycles.

Claim class: T4-H is now a **full THEOREM**, not a hypothesis or conditional claim. T1 ($N=2$ constructive) remains independently valid without invoking T4-H’s universal property. T4, T5 (K_joint Associativity) are upgraded accordingly.

10.2 Upgrade: 3-Observer Prediction from Class C-Conditional to Class C

The 3-observer registration transition mechanism (`3observer_registration_transition.md`, v1.1, 2026-05-28) was previously classified **Class C-conditional** pending T4-H Steps 3-4. With T4-H THEOREM confirmed, the conditional gate is resolved.

3-observer hierarchical configuration:

Hierarchy: $t_F1 < t_F2 < t_W$

```

F1 measures system S → registers k_F1 in K_{F1}
F2 measures F1's lab → registers k_F2 in K_{F2}
W (Superobserver) measures joint F1+F2 lab
  (requires_K_joint = 1; interference measurement)

```

The registration transition mechanism traces via four steps:

- **H1:** $K7_trace$ records $\Delta_closure(k_F1) = V_prov(k_F1) - V_final(k_F1)$ at $t_close(K_F1)$.
- **H2:** $K7_trace$ records $\Delta_closure(k_F2)$ similarly at $t_close(K_F2)$.
- **H3:** D_enc evaluates: $Enc(M_aware, k_F1) = 1$ iff the post-closure act M_aware 's outcome depends counterfactually on $\Delta_closure(k_F1)$. In the 3-OBS case, this propagation requires K_joint for $N=3$, provided by T4-H.
- **H4:** If $Enc = 1 \rightarrow requires_K_joint(M_aware, M_W) = 1 \rightarrow K5 \text{ fires} \rightarrow V(M_aware) \rightarrow 0 \rightarrow \text{no-awareness result propagates hierarchically}$.

Upgraded classification:

Component	Prior class	Current class	Condition
$K7_trace + D_enc$ application	C-conditional	Class C	T4-H THEOREM 2026-05-28
No-awareness propagation (H1-H4)	C-conditional	Class C	T4-H THEOREM 2026-05-28
Numerical prediction δ_M3	Illustrative	Illustrative	Conditional on $K9_E$ P9 only (see §10.3)

10.3 3-Observer Numerical Prediction (Illustrative)

Important scope boundary: The 3-observer *structural mechanism* (§10.2) is Class C — it follows from K-space axioms K1–K8 + Layer 2 ($K7_trace$, D_enc) + T4-H THEOREM. The 3-observer *numerical prediction* (δ_M3) requires the additional $K9_E$ postulate (P9) — it is illustrative and conditional on $K9_E$.

Prediction (conditional on $K9_E$ P9):

$\delta_M3 = -0.223$ at $\beta = 0.3$ (illustrative, not a measurement)

- 3-observer K_ctx is larger than 2-observer $K_ctx \rightarrow$ amplification factor $\approx 11\times$ relative to 2-observer δ_S .
- At $\beta = 0.3$, the 2-observer prediction gives $\delta_S \approx -0.020$; the 3-observer setup predicts $\delta_M3 \approx -0.223$.
- The amplification arises because $K_ctx(k_i, Exp)$ includes two contextual observer registrations instead of one, increasing f_perp .

Dependency chain for numerical prediction:

$K7_trace + D_enc$ (Layer 2, canonical)
 $\downarrow$ mechanism (Class C)
T4-H THEOREM ($N=3$ K_joint -- resolved)
 $\downarrow$ structural substrate
 $K9_E$ postulate P9 (Class C qualified -- not yet confirmed)
 $\downarrow$ probability assignment
 $\delta_M3 = -0.223$ at $\beta=0.3$ (illustrative)

Caveat: $\beta = 0.3$ is chosen for illustration (not from a fit to 3-observer data, which does not yet exist). The true β value, if $K9_E$ is correct, would be determined by a future 3-observer experiment. The $11\times$ amplification factor itself is structural (from K_ctx size) and holds for any non-zero β , not just $\beta = 0.3$.

11. φ Conjecture: Structure-Preserving Map $K \rightarrow B(H)$ (Class D Supporting)

This section is Class D supporting material — NOT the central claim of v3.0.

The central claims of this paper are the K-space architecture (§4), the K9_E postulate (§5), and the K9-S12 experimental specification (§8). The φ conjecture (Project B) is logically independent of K9_E (Project C) — see §5.6 for the independence statement.

11.1 The Conjecture and Its Motivation

Conjecture (Class D): There exists a structure-preserving map $\varphi: K \rightarrow B(H)$, where $B(H)$ is the algebra of bounded operators on Hilbert space H , such that φ encodes the registration-logic properties of K-space (K1–K8) in the operator-algebraic language of Standard QM.

Motivation: VVV-QMRF’s K-space axiomatizes *what gets registered* (registration-logic structure). Standard QM’s $B(H)$ axiomatizes *what can be measured* (operator-algebraic structure). If φ exists, it would formalize the correspondence between registration events in K and observable operators in $B(H)$ — not by identifying K with H , but by showing that K ’s registration-logic has a faithful structural image in operator-algebraic language.

Phase 1–4 completion (Project B Track B): The φ -map research program has completed four phases: - **Phase 1:** Target selection — $B(H)$ chosen as codomain with image $\text{Im}(\varphi) \subseteq \{P_o\} \cup \{0\}$ (projection sub-lattice). Three counter-arguments against $B(H)$ resolved. - **Phase 2:** Necessary conditions N_1 – N_T derived from K1–K8 and T1–T9 (§11.2). - **Phase 3:** φ -conditional interpretation scope analysis (§12.2). - **Phase 4:** Open items documented (φ -O2: N_6 sufficiency boundary confirmed as fundamental; C2 readiness 8.0/10).

Why $B(H)$ as codomain (not C^* -algebra, von Neumann algebra, or projection lattice):

Alternative	Problem
$P(H)$ projection lattice	Not an algebra — K6 authority-composition (φ -6) has no natural image
Von Neumann algebra M	Requires specifying which subalgebra — adds unwarranted structural choice
Functor to category C_{obs}	Loses concrete $B(H)$ operator intuition; abstract for current phase
$B(H)$ (chosen)	Contains all projections P_o ; admits φ -1 through φ -7’ conditions; standard QM home

EWF connection: In the EWF scenario, $K_F \perp_K K_W$ on the K-side maps to $[\iota(P_{\{o_F\}}), P_{\{o_W\}}] \neq 0$ in $B(H)$ — the familiar non-commutativity of quantum observables. The φ -conditional analysis (§12.2) re-frames each interpretation’s structural gap as the specific necessary condition N_i it lacks machinery to satisfy.

11.2 Necessary Conditions N_1 – N_T (Derived from K1–K8)

The following necessary conditions must hold for any map $\varphi: K \rightarrow B(H)$ to qualify as structure-preserving. These are necessary conditions — not sufficient for φ ’s existence.

Condition	Source axiom	Statement	B(H) encoding
φ -1 (Well-Definedness)	K1	$\varphi: K_R \rightarrow B(H)$ is a total function; $\forall k \in K_R, \varphi(k) \in B(H)$ defined	$\text{cert}(k) = 1$ for all $k \in K_R \rightarrow$ no undefined images
φ -2 (Lüders Order)	K2	$k_1 <_R k_2$ maps to Lüders update: $\rho \mapsto P\{o_2\} \cdot P_{-}\{o_1\} \cdot \rho \cdot P_{-}\{o_1\} \cdot P_{-}\{o_2\}$ (application order from temporal order)	Non-commutativity of $P_{-}\{o_1\}, P_{-}\{o_2\}$ mirrors non-symmetry of $<_R$
φ -3 (Cert-Reflection)	K3	$\varphi(k)$ is determined by k 's own tuple fields alone; for all $R' \neq R$: $\varphi_{-}R(k) \notin \varphi_{-}\{R'\}(k')$	$P_{-}o = o\rangle\langle o $ determined entirely by the registered outcome o — intrinsic
φ -4 (Validity-Positivity)	K4	$V(k) = 1 \rightarrow \varphi(k) = P_{-}o \geq 0, \varphi(k) \neq 0$; $V(k) = 0 \rightarrow \varphi(k) = 0$	Connects to AOE (Bong 2020): $V(k) = 1$ (default validity) $\rightarrow$ non-zero positive operator
φ -5 (Invalidation-Absorption)	K5	$V_{-}\text{final}(k) = 0$ (post-closure irreversible) $\rightarrow \varphi(k) = 0$; $P_{-}o \rightarrow 0$ is irreversible in $B(H)$	Zero operator is an absorbing element: $0 \cdot A = A \cdot 0 = 0$; no recovery without additional info
φ -6 (Authority-Composition)	K6	$\text{Auth}(k_2 \rightarrow k_1, C_K) = 1$ maps to $P_{-}\{o_2\} \cdot P_{-}\{o_1\} \neq 0$ (non-annihilating composition)	k_2 has authority over k_1 only if their projectors are not onto orthogonal subspaces
φ -7 (Embedding Naturality)	K8	Diagram commutes: $\varphi_{-}X(i(k)) = \iota(\varphi_{-}R(k))$ where $\iota: B(H_R) \rightarrow B(H_X)$ is tensor extension	$\iota(P_{-}o) = P_{-}o \otimes 1_{-}\{H_{-}\text{extra}\}$ — standard tensor product inclusion of subsystem projector
φ -7' (Closure Finalization)	K7	$\varphi_{-}\text{final}(k) = \varphi_{-}\text{prov}(k)$ at $t_{-}\text{close}$; post-closure φ is fixed; provisional $\varphi_{-}\text{prov}$ can change pre-closure	Fixes the operator image at $t_{-}\text{close}$ — after that, $\varphi(k) = 0$ or $P_{-}o$ cannot change

Condition	Source axiom	Statement	B(H) encoding
N_T (Theorem bridge)	T2, T3	$K_F \perp_K K_W$ (K-side) $\rightarrow$ $[\iota(P_{\{o_F\}}),$ $P_{\{o_W\}}] \neq 0$ (B(H)-side): non-commutativity as image of incommensurability	This is the φ -conditional EWF connection: operator non-commutativity is the B(H) image of K-side $\perp_K$

Derivation status: N_1 – N_T are derived under the assumption that φ exists. They state what φ *must* satisfy; they do not prove φ *does* exist. The relationship is: φ exists $\rightarrow$ N_1 – N_T hold. The converse (N_1 – N_T hold $\rightarrow$ φ exists) remains unproven. Open item φ -O2 (N_6 sufficiency for K6 authority-composition) is documented as a fundamental boundary.

11.3 $K \neq H$ Boundary Reaffirmed

A critical architectural boundary: φ is a map from K to $B(H)$, not an identification $K = H$.

Three distinct layers remain separate throughout the φ analysis: - **K (registration layer):** Tuples $\langle M, o, \text{cert}, t, V \rangle$ with registration-logic operations (K1–K8). Not a Hilbert space, not a probability space. - **ρ (physical state layer):** Density operators $\rho \in D(H)$, evolving under Standard QM (P1–P4). Not modified by VVV-QMRF. - **B(H) (observable algebra):** Bounded operators, containing projection operators P_o . The codomain of φ .

$\varphi: K \rightarrow B(H)$ maps registration events to operators, but it does not collapse K into H . The image $\text{Im}(\varphi) \subseteq \{P_o\} \cup \{0\}$ is a small subset of $B(H)$ — φ selects the operators relevant to registration from within the full operator algebra. This is the formal expression of $K \neq H$: K-side validity events map *into* $B(H)$ without being *identified with* H .

Why this matters: If $K = H$ were asserted, VVV-QMRF would be claiming that every registration event has a direct physical correspondent in the Hilbert space — this would constitute a modification of Standard QM. $K \neq H$ (preserved by φ) ensures VVV-QMRF remains a registration-layer analysis rather than a modification of quantum dynamics.

11.4 Class D Supporting Status

The φ conjecture is Class D: - The **existence** of φ is not proven. No constructive proof has been given that a map satisfying all of φ -1 through φ -7', N_T is well-defined for all K -spaces. - The necessary conditions are **not sufficient**. Satisfying N_1 – N_T would be required for φ to exist, but whether it is sufficient is an open mathematical question. - **No empirical prediction** derives solely from φ . K9_E (§5) provides all testable predictions independently of φ . - φ is **not falsified** by a K9_E null result. The φ conjecture concerns the formal relationship $K \rightarrow B(H)$, not the probability assignment $P(o|K)$.

The value of the φ analysis is interpretive: it provides the formal language for §12.2 (φ -conditional scope boundaries for existing QM interpretations) and maps the VVV-QMRF framework into a notation familiar to operator-algebraic quantum mechanics. It is supporting infrastructure for the project's long-term formalization goal, not its current testable claim.

12. Positioning Against Existing Interpretations

12.1 Architectural Comparison

Interpretation	Response to EWF paradox	VVV-QMRF difference
Copenhagen	WF is ill-posed; classical apparatus required	K3/K4 formalize why apparatus has registration authority: $\sigma_R(M) = 1$ intrinsically (no Heisenberg cut needed)
Many-Worlds	All outcomes occur; no paradox	Registration events are singular per K-side (K1 cert = 1 invariant); not globally branching
QBism	Facts are agent-relative; no paradox	Agrees on agent-relativity; adds formal K-side structure (K1–K8, six conditions) that QBism does not supply
Relational QM	Facts are relation-relative; no paradox	Closest existing framework; VVV-QMRF adds formal machinery (K3 self-cert, K7 closure, requires_K_joint) that RQM does not
Objective Collapse (GRW)	Physical collapse restores absolute facts	VVV-QMRF does not require physical collapse on ρ -side; K5 operates on K-side validity, independent of ρ dynamics

Relation to Relational Quantum Mechanics: Rovelli (1996) argues quantum states are relative to observers. VVV-QMRF adds the formal registration-layer structure explaining *why* they are relative: $\sigma_R(M)$ operates intrinsically within K_R , independently of $K_{\{R/\}}$. $K_F \perp_K K_W$ is the registration-layer account of Rovelli’s observer-relativity. This is a genuine extension: VVV-QMRF supplies the formal conditions (K3 self-cert, K2 temporal order, K5 invalidation, K7 closure, requires_K_joint predicate) that Relational QM does not.

12.2 φ -Conditional Scope Boundaries (Class D)

Section §11 derives necessary conditions N_1 – N_T for any structure-preserving map $\varphi: K \rightarrow B(H)$. These conditions re-frame each architectural gap in §12.1 as a **φ -conditional scope boundary**: the specific condition N_i that the interpretation lacks the structural machinery to satisfy.

Scope boundary convention: “Lacks the structural machinery for N_i ” means the interpretation does not supply the formal element required by N_i — not that the interpretation is incorrect within its own domain.

Interpretation	Architectural gap (§12.1)	φ -conditional scope boundary	N_i lacking
Copenhagen	No formal definition of classical apparatus	Cannot construct φ -domain element $k = \langle M, o, \text{cert}, t, V \rangle$ with cert field $\rightarrow \varphi$ has no defined domain	N₃ (cert-reflection, §11.2): $\text{cert} = \sigma_R(M)$ $= 1$ required to admit k into K_R
Many-Worlds	No physical observable distinguishes branches	No singular admission act per branch $\rightarrow$ K_R carrier set underspecified $\rightarrow \varphi$ total function (N_1) has no well-defined domain	N₁ (well-definedness, §11.2): φ total requires K_R well-defined; branching leaves carrier set underspecified
QBism	Subjective probability is not a physical quantity	No structural $V(k) \in \{0,1\}$ field: agent-relative belief is not a binary validity predicate $\rightarrow \varphi$ cannot satisfy validity-positivity	N₄ (validity-positivity, §11.2): $V(k) = 1 \rightarrow$ $\varphi(k) = P_o \geq 0$ requires V as structural field, not degree of belief
Relational QM	No temporal closure event	RQM has relational facts but no $t_close \rightarrow$ φ_final cannot be fixed	N₇ (closure-finalization, §11.2): $\varphi = \varphi_final$ fixed at t_close requires temporal closure boundary absent from RQM
Objective Collapse (GRW)	Collapse on ρ -side only	GRW collapse operates on ρ -side state; K -side $V: 1 \rightarrow 0$ irreversible transition not formally supplied	N₅ (invalidation- absorption, §11.2): K -side V -update and post-closure irreversibility not entailed by ρ -side collapse

Status — Class D: N_1 – N_T are derived under the assumption that $\varphi: K \rightarrow B(H)$ exists (§11). This §12.2 analysis means: *if φ were to exist, these are the structural elements existing interpretations would need to supply.* Neither φ 's existence nor the empirical correctness of these scope boundaries is asserted. See §13.4 for the relationship between this φ -conditional analysis and the broader architectural constraint framing.

12.3 Three Logically Independent Projects

VVV-QMRF comprises three projects related by a one-way motivation chain. This section makes that independence explicit to prevent the critique that Buddhist epistemology is post-hoc justification for a physics claim.

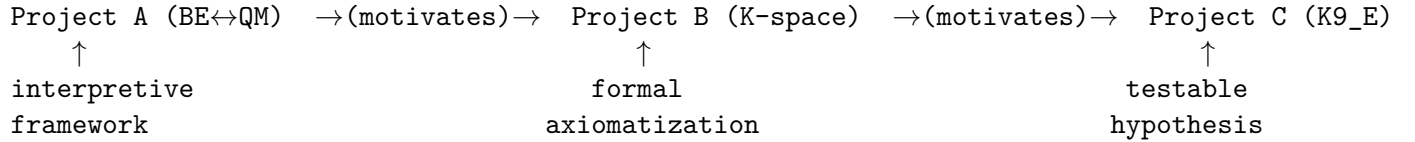
Project A — BE $\leftrightarrow$ QM Comparative Mapping (interpretive framework): - 30 nodes

(N_BE_00001–N_BE_00030), 39 edges (ED_BE_00001–ED_BE_00039) - Comparative framework between Buddhist Pramāṇa epistemology and quantum measurement concepts - Independently testable as a comparative-philosophy framework (internal consistency, historical accuracy of BE nodes, adequacy of QM node definitions) - **Does NOT constitute evidence for K-space axioms or K9_E**

Project B — VVV-QMRF Conceptual Framework (formal axiomatization): - K1–K8 frozen Layer 1 axioms; T1–T9 updatable Layer 2 bridge theorems; φ -map Class D conjecture - Independently testable as a formal axiom system (consistency, non-redundancy, expressive power) - Motivated by Project A’s structural analysis of valid registration, but not logically derived from it - **Does NOT make testable physical predictions by itself**

Project C — K9_E Testable Hypothesis (falsifiable claim): - K9_E postulate P9 with one free parameter β ; Class C (qualified) - Falsifiable via K9-S12 Modified Bong Protocol (§8); predicts $\delta\langle A_1 B_2 \rangle \neq 0$ at $\theta = 31^\circ$ - Motivated by Project B’s K-space structure ($K5_prospective \rightarrow T8 \rightarrow f_perp$), but K9_E is a *postulate* not *derivable* from K1–K8 - **This is the only project with a current falsification path**

A→B→C: one-way motivation, not logical derivation:



Each project is independently falsifiable: - Project A is falsified if the BE node definitions are historically inaccurate or the QM node definitions mischaracterize Standard QM. - Project B is falsified if K1–K8 are internally inconsistent, or if the K-space formalism cannot be applied consistently to any EWF scenario. - Project C is falsified if $\delta\langle AB \rangle = 0$ at $\theta = 31^\circ$ across a full θ -sweep (K9-S12 null result).

A null K9_E result ($\beta = 0$ measured by K9-S12) falsifies Project C but does not invalidate Project B (the K-space architecture remains valid as a formal structure) or Project A (the BE↔QM comparative framework stands independently).

13. Scope, Limitations, and Open Items

13.1 What This Paper Does Not Claim

- Does not replace Standard Quantum Mechanics or revise P1–P4.
- Does not revise the Born rule or unitary evolution (K9_E recovers Born exactly at $\beta = 0$).
- Does not claim Buddhist epistemology proves quantum mechanics (Project A is interpretive framework, not evidence).
- Does not claim K-side incommensurability is experimentally confirmed.
- Does not claim that consciousness plays any role in registration.
- Does not claim the EWF paradox is fully resolved on the ρ -side.
- Does not claim requires_K_joint is a necessary-and-sufficient condition for all LF violations; it is proposed only as a necessary registration-layer condition.
- **Does not claim K9_E is experimentally confirmed** — K9_E is Class C (qualified): structurally testable, empirically unconfirmed. The 2.31σ Proietti fit is real but ambiguous;

noise cannot be ruled out as an alternative explanation (§7).

- **Does not claim the φ -map $\varphi: \mathbf{K} \rightarrow \mathbf{B}(\mathbf{H})$ is proven to exist** — the map is conjectured (Class D). The necessary conditions N_1 – N_T are derived but not sufficient for φ ’s existence (§11).
- **Does not claim $K9_E$ is the unique probability rule consistent with $K1$ – $K8$** — $K1$ – $K8$ are structural axioms that do not uniquely determine a probability rule; $K9_E$ is one motivated postulate, not the only possibility.
- **Does not claim the 3-observer prediction $\delta_M3 = -0.223$ is independent of β** — this value is illustrative at $\beta = 0.3$; the true β is unknown until experimental measurement.

13.2 Formal Phase Summary and Open Items

The $\perp_K$ formal definition chain is complete at the proposed Class D/C level:

Formal layer	Symbol	Class	Defined in
K-side incommensurability	$\perp_K$	D	§6.4
Admissible joint K-space	K_joint / $AdmJoint$	D	§6.3
Joint-validity demand	D_joint / $requires_K_joint$	D	§6.3
K-side comparison context	C_K	D	§6.4
Bridge lemma	$Bridge_EWF$	D/C	§6.5
Operational data criterion	ODC_K	C	§6.6
K-Space axioms (Layer 1)	$K1$ – $K8$	D ($K1=C$)	§4.1–4.2
Bridge theorems (Layer 2)	$T1$ – $T9$, $K7_trace$, D_enc	D/C	§4.3–4.4
Probability postulate	$K9_E$ (P9)	C (qualified)	§5
Experimental protocol	$K9$ -S12	C	§8
φ conjecture	$\varphi: \mathbf{K} \rightarrow \mathbf{B}(\mathbf{H})$	D (supporting)	§11

Resolved items (updated from v2.0):

Item	v2.0 status	v3.0 status
Axiomatize K as full mathematical structure	Deferred long-term task	RESOLVED — $K1$ – $K8$ v2.4, $T1$ – $T9$, $K7_trace$, D_enc (§4)
$K5_prospective$ (A1 assumption)	Class D semantic extension	RESOLVED — upgraded to $K5_prospective$ clause (§4.2)
$T4$ -H colimit existence	Conditional hypothesis	THEOREM (4/4 steps, RCA 4.74/5, 2026-05-28, §10.1)
3-OBS Class C-conditional	Conditional on $T4$ -H	Class C ($T4$ -H resolved, §10.2)
$K7_trace$ canonical status	BB-VVV local construct	Canonical Layer 2 (4 consumers, RCA 4.77/5, §4.4)

Deferred proof items (not required for current claim class):

Deferred item	Reason deferred
Full semantic proof for Bridge_EWF	Requires formal semantics of “no admissible reinterpretation”; paper uses operational sufficient conditions
Full formal proof for $\perp_K$ as mathematical relation	K1–K8 structural definition complete; full topological/order-theoretic treatment deferred
AdmJoint necessary-and-sufficient conditions	Currently sufficient conditions A–E; full characterization is an open item
Equivalence of $\sigma(M)$ and $R^{\wedge}_{\text{svasa}}$ formalisms	Separate research track

13.3 Falsification Condition (Restated)

K9_E (Project C) is falsified by: a K9-S12 experimental result with $\delta\langle A_1 B_2 \rangle = 0$ at $\theta = 31^\circ$ within predeclared statistical tolerance across a full θ -sweep. This would falsify the overlap-only class of deformations (the class that K9_E belongs to), not just a specific β value.

The K-side incommensurability conjecture is falsified by: an EWF experiment producing results consistent with a single K_joint registration model satisfying Conditions 1–6 for both observers simultaneously, for a configuration independently classified as $\text{requires_K_joint} = 1$ via D_joint and Bridge_EWF. If K_joint exists empirically, the Bridge_EWF conditional bridge requires revision.

K9_E falsification (K9-S12 null result) does not falsify: - The K-space architectural framework (K1–K8 + Layer 2) — Project B - The K-side incommensurability conjecture ($K_F \perp_K K_W$) — a different structural claim - The $BE \leftrightarrow QM$ comparative mapping — Project A

Operationally, all three falsification tests require a predefined data criterion (tolerance τ fixed before data collection); otherwise the criterion becomes post-hoc.

13.4 Architectural Constraints of the K-side/ ρ -side Separation

The core architectural commitment of VVV-QMRF is $K \neq H$: the registration layer is structurally distinct from the physical layer (§2.1). This separation enables the framework to address the registration layer gap. It also imposes two architectural constraints that are inherent consequences of the layer separation, not oversights.

Constraint 1 — Self-certification is not a ρ -side observable. Condition 4 requires $\sigma_R(M) = 1$ to be determined intrinsically within K_R . Any physical measurement of σ would itself require a second-order registration act, re-entering the von Neumann regress that K3 is designed to terminate. The operational compromise is stated in ODC_K condition (ii): self-certification is treated as a structural postulate satisfied by construction when the registering system meets the architectural definition of a K-side observer.

Constraint 2 — D_joint is a framework-level classification, not a physical observable. The predicate $D_joint(A, B, Arch)$ classifies whether a comparison architecture demands joint validity. Operational sufficient conditions A–E (§6.3) connect observable features of the setup (interference vs. readout, comparison vs. no comparison) to D_joint values in a reproducible way. But the mapping is a VVV-QMRF classification of the experimental architecture, not a measurement of a physical quantity.

These constraints are typical of the measurement problem, not unique to VVV-QMRF. Every interpretation or extension of quantum mechanics that addresses measurement introduces an unobservable primitive:

Framework	Central concept	Why not directly observable
Copenhagen	Classical apparatus / Heisenberg cut	No formal definition of what makes an apparatus classical
Many-Worlds (Everett)	Branching of worlds	No physical observable distinguishes one branch
QBism	Agent’s degree of belief	Subjective probability is not a physical quantity
Relational QM (Rovelli)	Facts relative to observer	Relata are only defined within the relation
Objective Collapse (GRW)	Collapse mechanism	Collapse events are postulated, not yet detected
VVV-QMRF	Self-certification / D_{joint}	K-side properties are not ρ-side observables ($K \neq H$)

What distinguishes VVV-QMRF is that the constraints are explicitly named, traced to a single architectural commitment ($K \neq H$), and given operational bridges (ODC_K, conditions A–E) that define what can and cannot be tested under the current formalism.

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Appendices

Appendix A — K9_E reproduction scripts: `documents/research_documents/project_vvv_qmrf_class_c/07_f`
Appendix B — K_Space_Axiomatization v2.4: `documents/research_documents/meta_architecture/K_Space`
Appendix C — AHP audit footprint: `documents/research_documents/anti_hallucinations/`

Draft phase status

Phase	Sections	Status
P1	Skeleton + Abstract (8 paras) + §4 K-Space Architecture	COMPLETE 2026-05-28
P2	§5 K9_E Postulate + §7 Empirical Status	COMPLETE 2026-05-28
P3	§8 K9-S12 + §10 T4-H + 3-OBS	COMPLETE 2026-05-28
P4	§9 BB+FR + §11 φ -map Class D	COMPLETE 2026-05-28
P5	Carry-over §1, §2, §3 (headers) + §6 + §12 + §13	COMPLETE 2026-05-28
P6	Abstract finalization + header update	COMPLETE 2026-05-28
P7	CHANGELOG + README + AHP footprint verified	COMPLETE 2026-05-28

VVV-QMRF Working Paper v3.0 — All phases P1-P7 complete (2026-05-28). All mini-RCA PASS (4.57–4.6/5). Average 4.58/5. Promoted from draft to final 2026-05-28. Abstract (8 para) + §1–§13 (~14,500w). Supporting files: CHANGELOG.md, README.md, plan. References [13] and [14] finalized; [14] arXiv ID pending confirmation post-submission. All formal claims classified by evidence level. Class C = structurally testable, empirically UNCONFIRMED. Class D = proposed.